



Igor Slazyk

Machine Learning Engineer

Portfolio

islazykv.github.io

LinkedIn

linkedin.com/in/igor-slazyk

GitHub

github.com/islazykv

Personal Information

Nationality: Polish
Date of Birth: 03.11.1992
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Profile

I am a positive and cheerful person who enjoys challenges. I approach every task with dedication and commitment, seeking effective solutions and paying attention to details. I am communicative and team-playing oriented, but I can also work independently and take initiative. I am open-minded, adapt quickly and I am always eager to learn.

Interests

Popular Science Literature
IT / Technology
Gaming
Cycling
Hiking

Summary

ML engineer with 8 years of experience in data science, including 4 years focused on machine learning. Passionate about turning data into valuable insights and impactful solutions. Driven to produce high-quality, scalable code and models. Collaborative team player, eager to contribute to projects at the intersection of AI and MLOps.

Education

- **Ph.D. in Particle Physics** 10.2017 – 07.2021
University of Ferrara Ferrara, Italy
Institute of Nuclear Physics Krakow, Poland
- **M.Sc. in Applied Physics** 02.2016 – 05.2017
Krakow University of Technology Krakow, Poland
- **B.Sc. in Applied Physics** 09.2012 – 02.2016
Krakow University of Technology Krakow, Poland
- **B.Sc. in Civil Engineering** 09.2011 – 02.2015
Krakow University of Technology Krakow, Poland

Experience

- **Postdoctoral Research Fellow** 01.2022 – currently
Høgskulen på Vestlandet Bergen, Norway
 - Development of ML models for data analysis in high-energy physics
 - Design and implementation of scalable end-to-end ML pipelines
 - Data preprocessing, feature engineering, model evaluation and optimization
 - Data analysis and interpretation of results from ML-based physics searches
 - Code maintenance, refactoring and automated testing
 - Preparation of publications and presentation of ML results
- **Affiliated Researcher (ATLAS)** 01.2022 – 12.2025
CERN Geneve, Switzerland
 - Software development within the ATLAS experiment framework (Athena)
 - Development of TauCP software for tau particle reconstruction and identification
 - Code optimization, refactoring and integration into large-scale workflows
 - Evaluation and interpretation of physics analysis results
 - Collaboration with developers, analysts and physicists on software and data tools
- **Affiliated Researcher (LHCb)** 12.2017 – 03.2021
CERN Geneve, Switzerland
 - Software development for automated quality assurance of RICH detector modules
 - Data analysis of quality assurance results for elementary detector cells
 - Development of data processing scripts and tools for measurement workflows
 - Hardware assembly, testing and quality control measurements
 - Initial exploration of ML approaches for detector data analysis
- **Front-End Developer** 10.2016 – 09.2017
Freelance Krakow, Poland
 - Development of responsive websites using React.js
 - Collaboration with back-end developers on feature integration and functionality
- **R&D Physics Intern** 04.2016 – 09.2016
Camlin Technologies Zurich, Switzerland
 - Data analysis of optical and quantum measurements of semiconductor materials
 - Data collection and processing for photoluminescence spectroscopy
 - Collaboration with software developers and physicists on analysis workflows

Projects	
repository notebook showcase publication	Tau Supersymmetry Search Boosted Decision Trees Neural Networks - Developed a supervised ML pipeline for multiclass classification with MLflow - Trained and optimized classifiers with Optuna tuning - Implemented ML-based regions construction and statistical hypothesis testing
	Tau Anomaly Detection Autoencoder Variational Autoencoder - Developed an unsupervised ML pipeline for anomaly detection with W&B - Trained and optimized models with Ray Tune tuning - Implemented reconstruction error anomaly scoring and latent space analysis
	Tau Energy Scale Mixture Density Network Neural Networks - Developed a probabilistic regression model with uncertainty estimation - Trained with Huber loss pretraining and NLL fine-tuning for density estimation - Evaluated model performance with energy response and resolution histograms
publication	Sphalerons and Black Holes Convolutional Neural Network Boosted Decision Trees - Optimized high-capacity models to classify rare high-multiplicity LHC events - Developed a pipeline mapping detector signals into three-layer event images - Compared high-level features with raw pixel data to maximize signal sensitivity
	Di-Tau Reconstruction Software Development Data Science - Engineered a C++ analysis framework for boosted di-tau reconstruction studies - Benchmarked fixed & variable radius subjects to optimize reconstruction efficiency - Evaluated diverse seed jet collections to establish baseline configurations
	Elementary Cells Software Development Data Science - Developed an automated C++/Python/LabVIEW software for photomultiplier QA - Analyzed array performance via noise, dark count and threshold measurements - Explored ML-based particle identification for the LHCb RICH detector

Skills									
Languages <table><tr><td>English:</td><td>Professional</td></tr><tr><td>Norwegian:</td><td>Basic</td></tr><tr><td>Italian:</td><td>Basic</td></tr><tr><td>Polish:</td><td>Native</td></tr></table> Currently Learning: - Kubernetes - Azure ML - Azure DevOps Pipelines - Databricks	English:	Professional	Norwegian:	Basic	Italian:	Basic	Polish:	Native	Programming Languages & Tools: - Python, C++, SQL, Bash - Git, Jupyter, pytest, uv, conda Data Processing & Visualization: - NumPy, pandas, SciPy, ROOT, uproot, Awkward Array, pyhf - Matplotlib, seaborn, Plotly Machine Learning & Deep Learning: - scikit-learn, XGBoost - PyTorch, PyTorch Lightning, TensorFlow, Keras Tuning & Explainability: - Optuna, Ray Tune, SHAP Tracking & Deployment: - MLflow, W&B, DVC, Hydra - FastAPI, Docker, GitHub Actions Web Development: - HTML, CSS, JavaScript, React
English:	Professional								
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